

Agenda

- 1st day:
 - Overview & Planning & Installation
 - Architecture & System Configuration
 - Licensing
 - Application Configuration
 - Journaling
 - Backup & Restore



Agenda

- 2nd day:
 - Managing Database
 - Managing Process
 - Task & Automation
 - Mirroring
 - Security
 - Monitoring & Debug



Overview



Overview

- InterSystems IRIS® data platform.
- InterSystems IRIS for Health™.
- Tools.
- Platforms and versions.
- Documentation.



InterSystems IRIS

- Complete high-performance data platform combining multi-model database access, interoperability and analytics.
- Includes:
 - InterSystems IRIS® Database.
 - Formerly Caché.
 - InterSystems IRIS® Interoperability.
 - Formerly Ensemble.
 - InterSystems IRIS® Business Intelligence.
 - Formerly DeepSee.



InterSystems IRIS (cont.)

- Multiple data models.
 - Object, relational, document, vector, multidimensional.
 - Relational model includes columnar storage options.
- Development languages.
 - InterSystems ObjectScript.
 - Embedded Python.
 - SQL.
- External languages, APIs, and protocols.
 - Python, Java, .NET, Node.js.
 - ODBC/JDBC.
 - REST and SOAP.



InterSystems IRIS (cont.)

- System Interoperability.
 - Message-based architecture for orchestration among multiple systems.
 - Includes data routing, transformations and workflows.
- Hybrid transaction/analytical processing (HTAP).
 - Includes business intelligence and machine learning capabilities.



InterSystems IRIS (cont.)

- Horizontal scaling.
 - Distributed caching for user volume (ECP).
 - Sharding for data volume.
- Easily provision and deploy InterSystems IRIS in containers on public or private clouds using:
 - InterSystems Kubernetes Operator.
 - InterSystems Cloud Manager.



InterSystems IRIS for Health

- Data platform with out-of-the-box tools for building healthcare solutions.
 - Support for major healthcare standards including HL7[®] FHIR[®].
 - Interoperability tools.
 - Pre-built extensible transformations between healthcare messaging formats.
- Built on InterSystems IRIS data platform.



Tools

- Integrated development environments (IDEs).
 - Visual Studio Code.
 - InterSystems Studio (Windows only).
- ObjectScript, Python, and SQL shells.
 - InterSystems Terminal (Windows only).
 - Launched from any terminal emulator.
- Management Portal.
 - Browser-based tools.
- All tools can access local or remote installations.



InterSystems IRIS and IRIS for Health Platforms and Versions

- [Version Information page.](#)
- Platforms:
 - UNIX®, Linux, Windows.
 - Docker container on Linux.
- Release schedule:

Release	Numbering	Notes
Extended Maintenance (EM)	20# #.1. <i>n</i>	Released at beginning of year. Includes maintenance releases during year.
Continuous Delivery (CD)	20# #.2.0 20# #.3.0 20# #. <i>n</i> .0	Early access to new features to be included in next year's EM release. No maintenance releases.



Documentation

- Located at docs.intersystems.com.
- Available as web pages or PDF.



Planning

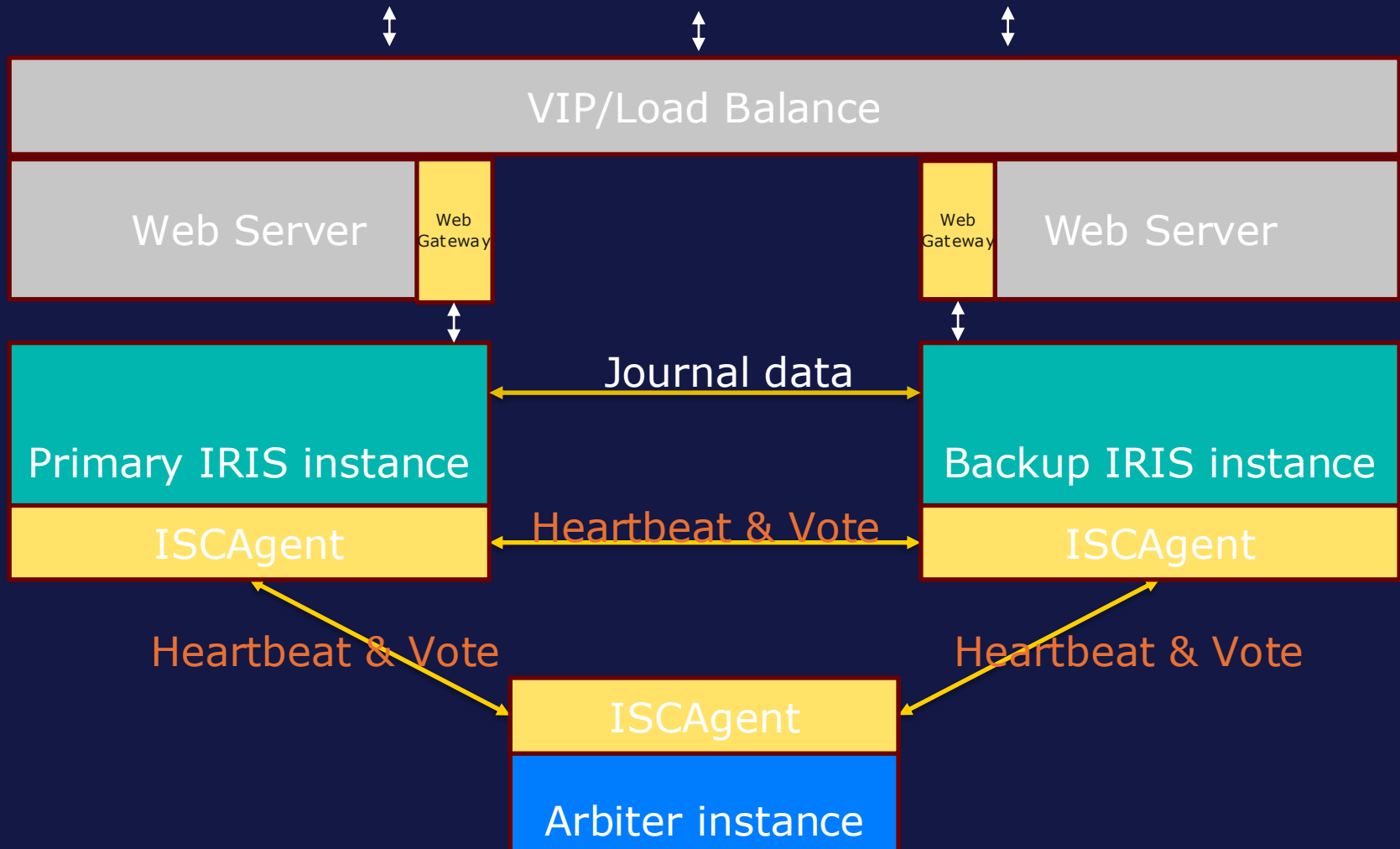


Planning

- Network
 - Hostname/IP
 - Firewall/Ports
 - VIP/Load balance
- Servers
 - CPU cores
 - Memory
 - Storage/File Folders
 - Components/Install kits



Planning



Folders



IRIS folders

- Internal folders
 - IRIS Install Dir
 - IRIS Data files Dir
 - IRIS Journal files Dir
 - IRIS Log files Dir
 - IRIS Registry Dir – the folder for registry information on Linux
- Change folder place for performance & safety
 - Put journal file to isolated disk
- External folders – which IRIS may need to access
 - The folder containing certification files
 - The folder containing other files required by IRIS instance, iscagent processes or External Language Server processes, like jar files



Permission Checker: filecheck

- Validate permissions on files at instance-startup or on demand
- 1st Phase scan: IRIS installation directory and subdirectories (bin, mgr, csp, csp/bin)
 - Using the IRIS registry
- 2nd Phase scan: Database, Stream, Journal and WIJ directories
 - Using the CPF file



Ports

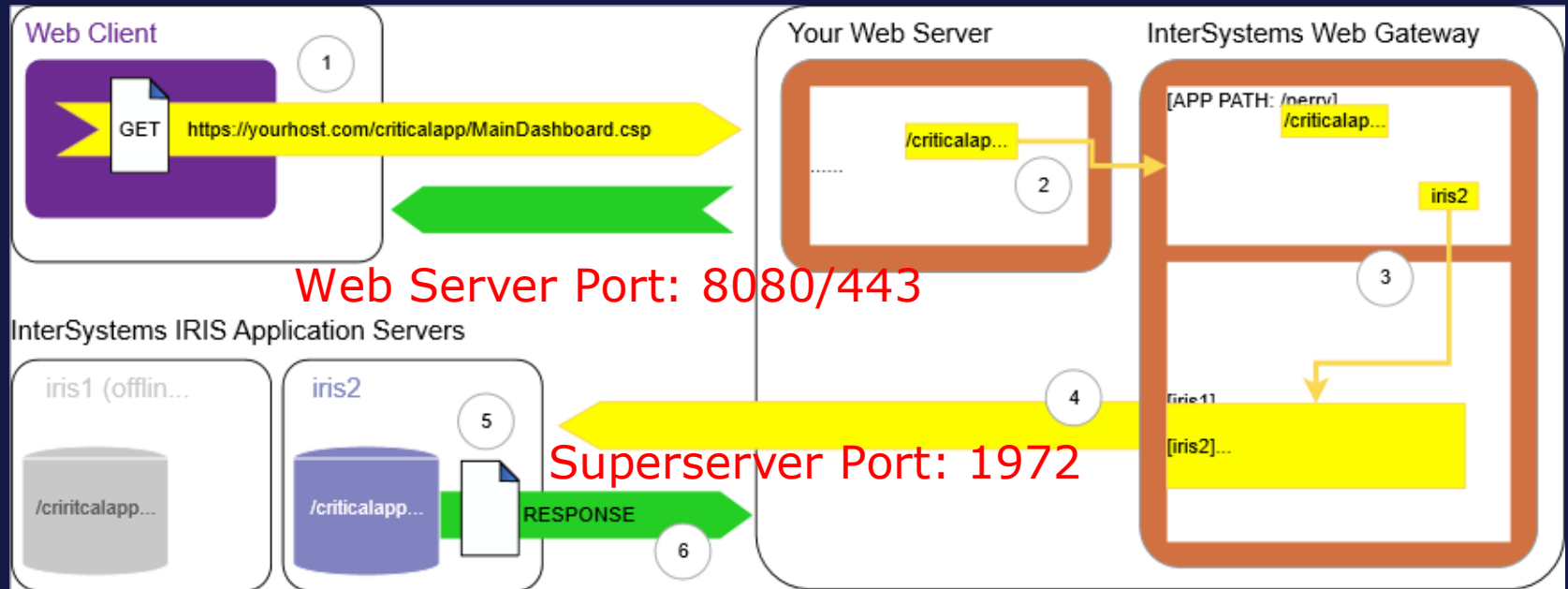


Ports

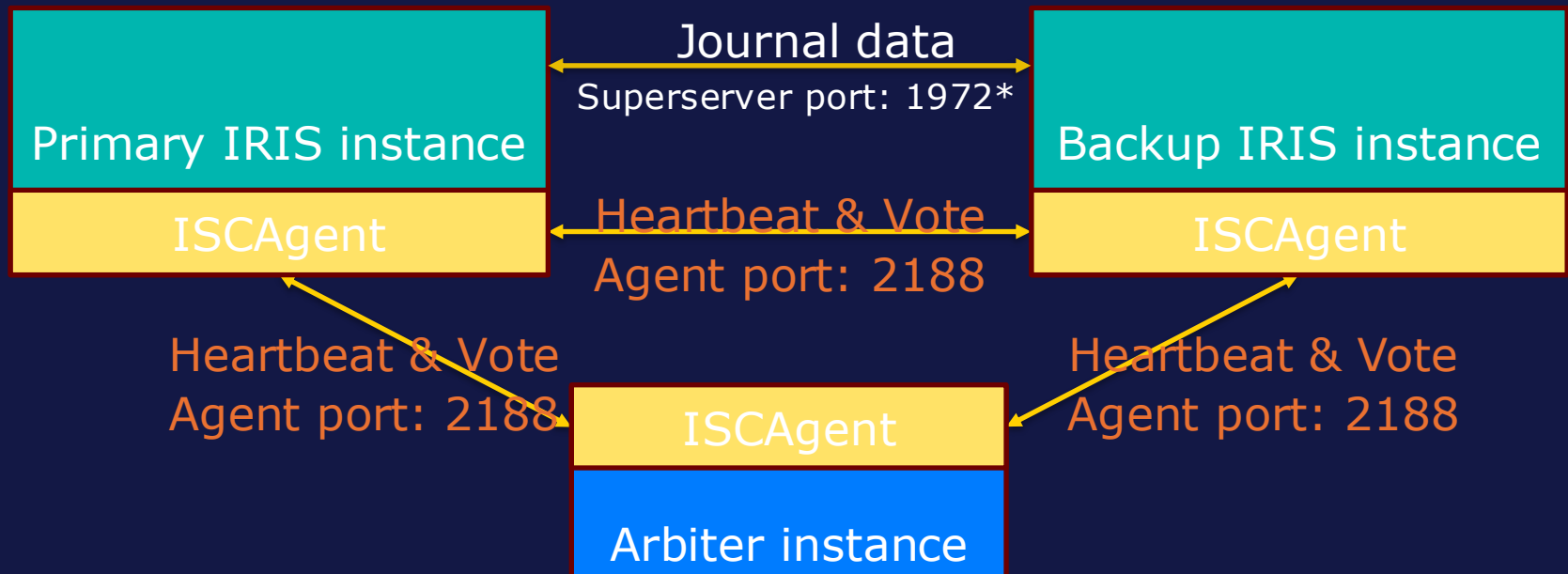
- Superserver port:
 - All traffic between application clients and IRIS, including ECP, Mirror, Sharding cluster
 - Default is 1972
- Web Server port:
 - HTTP based traffic will go through web server port
 - HTTP, HTTPS, SOAP, RESTful
 - VSCode
 - Web Server port is defined by the web server. Default is 80, 443
- Agent port:
 - Traffic on heartbeat and vote among ISCAgent in mirroring cluster
- Other ports:
 - License server – Default is 4002
 - External Language Servers, SQL Gateway
 - SSH (22)



Superserver Port and Web Server Port



Agent Port - Mirroring cluster



*Notice:

1. Only user databases are mirrored, not system databases nor temp database, not file system
2. Under some situation, journal data may send by ISCAgent through Agent port



Other ports – External Language Servers

- External Language Server ports used by InterSystems IRIS are:
 - Bound to localhost (127.0.0.1), not 0.0.0.0
 - Usually not exposed to external networks
 - Used only for inter-process communication between IRIS and local runtime (Java/Python/.NET)
 - These external language server processes are started by user irisusr
- Therefore:
 - No external attack surface exists
 - Ports are not reachable from outside the host
 - This complies with least privilege and secure design principles

System > Configuration > External Language Servers

External Language Servers

Create External Language Server

List of currently defined External Language Server definitions:

Filter: Page size: 0 Max rows: 1000 Results: 8 Page: 1 of 1

Name	Type	Port	Activity Log	Start	Stop	●●●	Delete
%DotNet Server	.NET	53372	Activity Log	Start	Stop	●●●	Delete
%IntegratedML Server	ML	53572	Activity Log	Start	Stop	●●●	Delete
%JDBC Server	JDBC	53772	Activity Log	Start	Stop	●●●	Delete
%Java Server	Java	53272	Activity Log	Start	Stop	●●●	Delete
%ODBC Server	ODBC	53972	Activity Log	Start	Stop	●●●	Delete
%Python Server	Python	53472	Activity Log	Start	Stop	●●●	Delete
%R Server	R	53872	Activity Log	Start	Stop	●●●	Delete
%XSLT Server	XSLT	53672	Activity Log	Start	Stop	●●●	Delete



Memory



Memory Planning

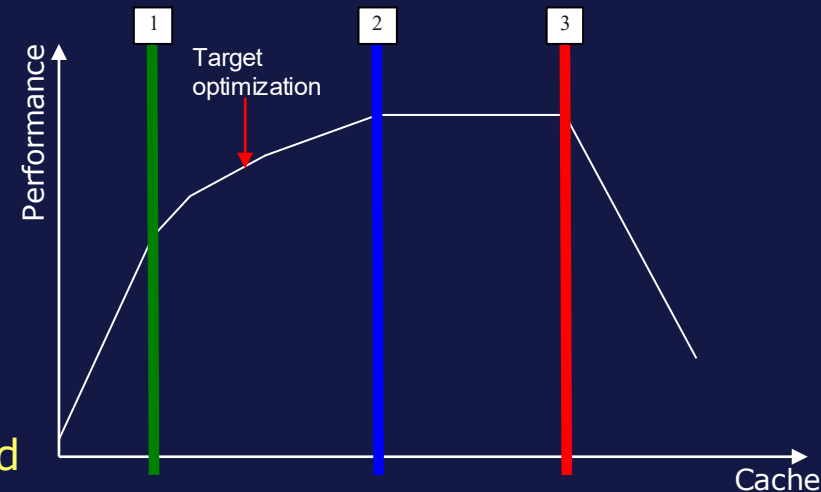
- Initial memory planning for regular setup
 - Keep memory/cores ratio to 8:1
 - for example, have 64GB memory for 8 cores server



Memory Planning – IRIS side

■ IRIS cache/memory planning (Shared Memory)

- Database cache (for data):
 - 32MB-16TB
- Routine cache (for logic/code):
 - 36MB-64GB
- gmheap (shared memory heap):
 - such as lock tables, global mapping, database name and directory information, the security system, and so on
 - 16384-1,073,741,760 (KB)
- Journal Buffers (jrnbufs):
 - For journal data
 - Default value is 64 MB



■ IRIS resource planning



Memory Planning – OS side

- **Large page/Huge page (/etc/sysctl.conf)**
 - The amount of memory available in huge pages should be greater than the total amount of shared memory to be allocated; if it is not greater, huge pages are not used.
 - For example, with 3580 MB of shared memory and a huge page size of 2 MB, at least 1791 huge pages should be configured.
- **Dirty memory page (/etc/sysctl.conf)**
 - Lower dirty memory setting to make I/O stable and let IRIS control I/O
 - `vm.dirty_background_ratio = 5`
`vm.dirty_ratio = 10`



Installation



Overview

- Installation.
- Starting/stopping instance.
- Calling utilities.
- Emergency access.
- Multiple instances.



Installation

- Can install multiple instances on same server, on premises or in cloud.
- Installation process prompts for:
 - Unique name for instance.
 - Unique installation directory.
 - Type of installation.
 - Automatic configuration of installed local web server.



Minimum CPU Microarchitecture

- Pre-v2024.1, installation did not check CPU microarchitecture.
- In v2024.1+, InterSystems introduced minimum supported CPU microarchitecture requirements.
 - Installation stops if requirements not met.

	Intel	AMD
v2024.1	Haswell	Bulldozer
v2025.3+	Skylake	Excavator



Types of Installation

- Development
 - Installs all server and client components.
 - Studio (Windows only) installed pre-v2024.2.
 - Installs Web Gateway.
- Server
 - Installs server components and Web Gateway.
- Client
 - Installs Java/.Net bindings and xDBC drivers.
 - On Windows, installs Studio (pre-v2024.2) and Launcher.
 - No support for local databases.



Types of Installation (cont.)

- Web Server
 - Installs Web Gateway on Web Server machine.
- Custom
 - Can select which components to install.



Upgrading

- Installer provides option to upgrade existing instances.
 - Upgrade removes Studio. (v2024.2+)
 - After upgrade, can install v2024.1 Studio as separate component.
 - Backwards and forwards compatible.
 - Future IRIS upgrades will not remove separate Studio component.
- Refer to:
 - *Installation Kit > Upgrading and Modifying.*
 - Upgrade Impact Checklist
(docs.intersystems.com/upgrade)



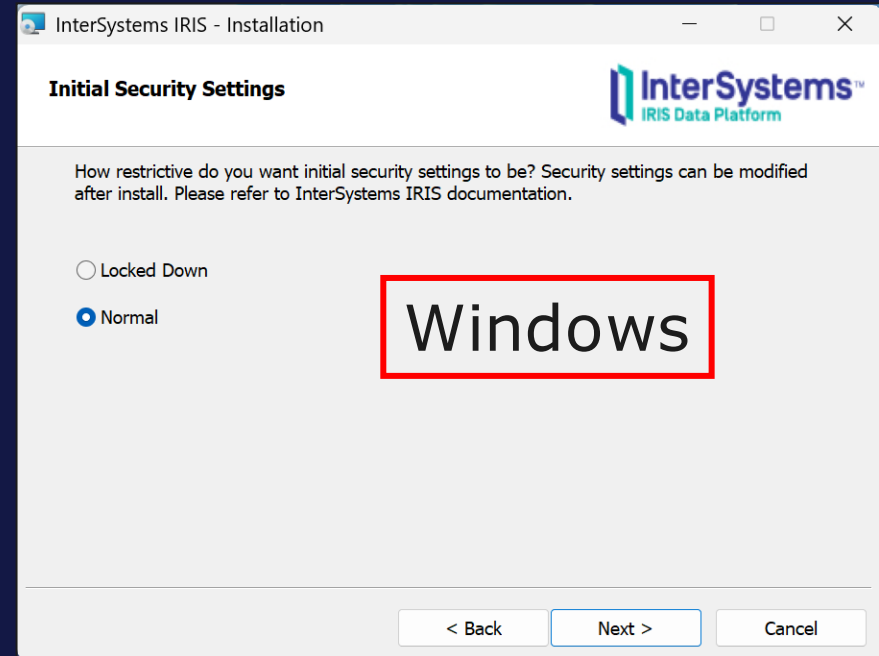
Unicode or 8-bit Support

- Unicode support handles characters in Unicode (16-bit) format.
 - For applications using languages that store data in Unicode format, such as Thai.
- 8-bit support handles characters in 8-bit format.
- Cannot revert from Unicode to 8-bit support.



Initial Security Configuration

- Choose between Normal and Locked Down.
- Some products also provide Minimal option.
 - Allows unauthenticated access and provides full privileges.
 - Not recommended.



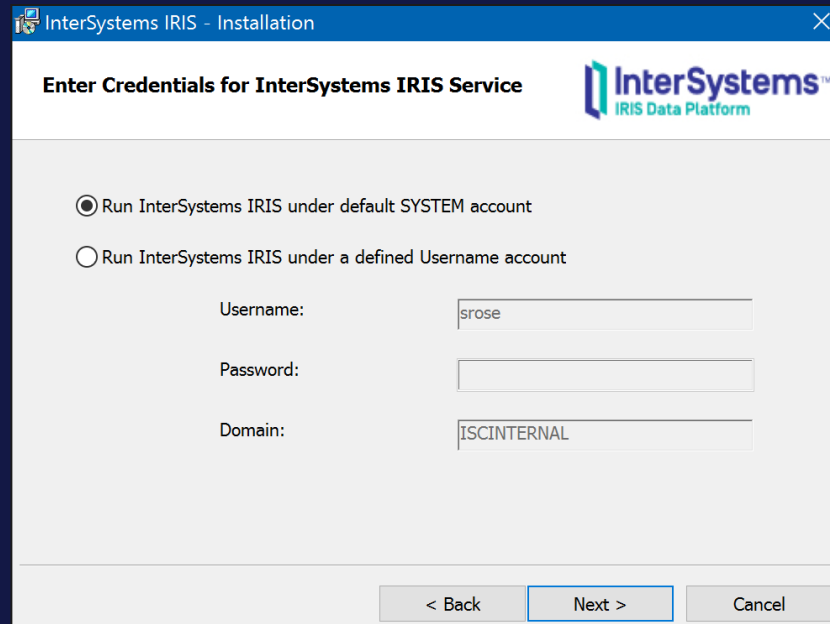
```
How restrictive do you want the initial Security settings to be?
"Locked Down" is the most secure, "Normal" is the least restrictive.
1) Locked Down
2) Normal
Initial Security settings <1>? 
```

Unix® and Linux



Windows Installation Notes

- During Normal and Locked Down install, asks for OS account under which to run instance.
 - Recommend using separate user account.



The screenshot shows a Windows installation window titled "InterSystems IRIS - Installation". The main heading is "Enter Credentials for InterSystems IRIS Service". The InterSystems IRIS Data Platform logo is in the top right. There are two radio button options: "Run InterSystems IRIS under default SYSTEM account" (which is selected) and "Run InterSystems IRIS under a defined Username account". Below these are three text input fields: "Username:" with the value "srose", "Password:" (empty), and "Domain:" with the value "ISCINTERNAL". At the bottom are three buttons: "< Back", "Next >" (highlighted with a blue border), and "Cancel".



UNIX[®] and Linux Installation Notes

- Run install using sudo command.
 - Allows limited non-root installs when sudo not available.
- During Normal and Locked Down install, asks for instance owner.
 - Enter OS user with administrator privileges.
 - For Minimal, root is instance owner.
 - Not recommended.
- All installs ask for group that can start/stop instance:
 - Enter group that includes administrators.



New User

- During Normal and Locked Down install, creates new user definition.
 - Windows: Same as OS user who did installation.
 - UNIX® and Linux: Same as instance owner.

New user



Name	Full Name
Admin	System Administrator
CSPSystem	Web Gateway user
IAM	User for /api/iam Web Application
srose	User who installed system
SuperUser	System Super user
UnknownUser	Unauthenticated user
Ensemble	Interoperability Manager (Internal use - not for login)
PUBLIC	(Internal use - not for login)
SYSTEM	SQL System Manager



UNIX® and Linux Installation

- Installer creates group called irisusr if it does not exist.
- For installed files:
 - Most owned by instance owner and irisusr group.
- When instance starts:
 - Sets its process groupid to irisusr.
 - IRIS system processes run as irisusr.
 - Daemons run as instance owner.



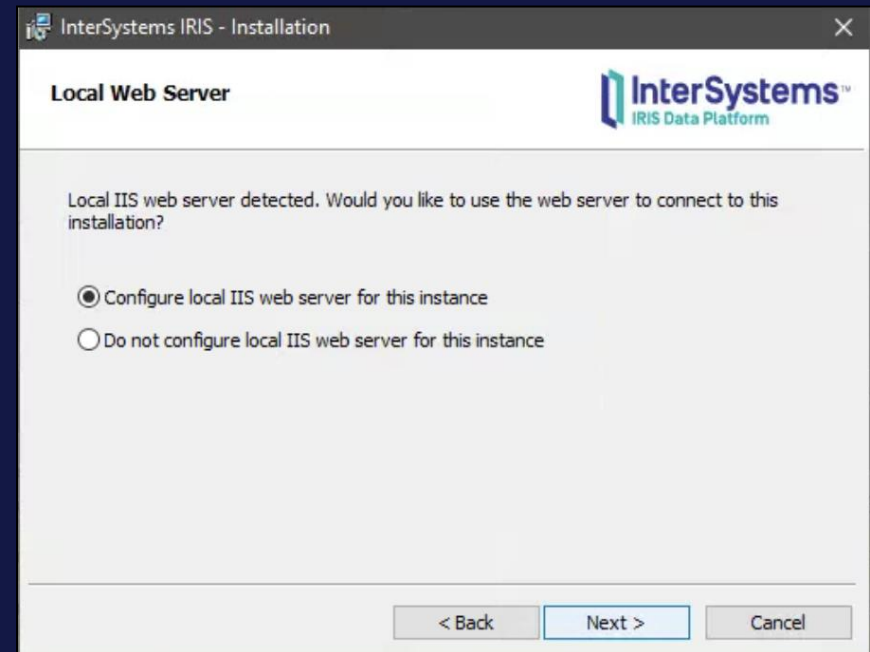
IRIS Requires Web Server

- Used for:
 - Management Portal.
 - Web Applications.
- Web server can be local or remote.
- Can manually configure supported web servers for IRIS:
 - Microsoft Internet Information Services (IIS).
 - Apache.
 - Nginx.



Automatic IIS & Apache Configuration

- Installer asks to configure local web server, if found.
 - After installation, verify settings for live environment.
- Web server then can connect to IRIS via Web Gateway.
- Installer does **not** install any web server.



Port Numbers Assigned During Installation

- Unique Super Server port number.
 - First installation: 1972.
 - Additional installations: 51773, 51774, and so on.
- Web Server port number.
 - Typically port 80 or 8080.
 - Multiple installations on same server typically share web server.
- Non-unique License Server port number.
 - Refer to Licensing module for more information.



Changing Port Numbers After Installation

- For Super Server and Web Server, in System Administration > Configuration:
 - System Configuration > Memory and Startup > Superserver Port Number.
 - Additional Settings > Startup > WebServerPort.
- For License Server, in System Administration > Licensing:
 - License Servers > click License Server name.



Multiple Superserver Ports (v2024.1+)

- Can add additional superserver ports for controlling different connection types.
 - System Administration > Security > Superservers.
- Must use default superserver port for Distributed Caching and Mirroring connections.

System > Security Management > Superservers

Superservers

Create New Superserver

Last update: 2024-03-27 10:44:10.361

The following is a list of superserver definitions:

Filter: Page size: 0 Max rows: 1000 Results: 1 Page: < << 1 >> > of 1

Port	Bind Address	Enabled	System Default
51775		Yes	Yes



Startup Actions

- Upon startup, instance performs many actions.
 - Review most actions in System Operation > System Logs > Messages Log.

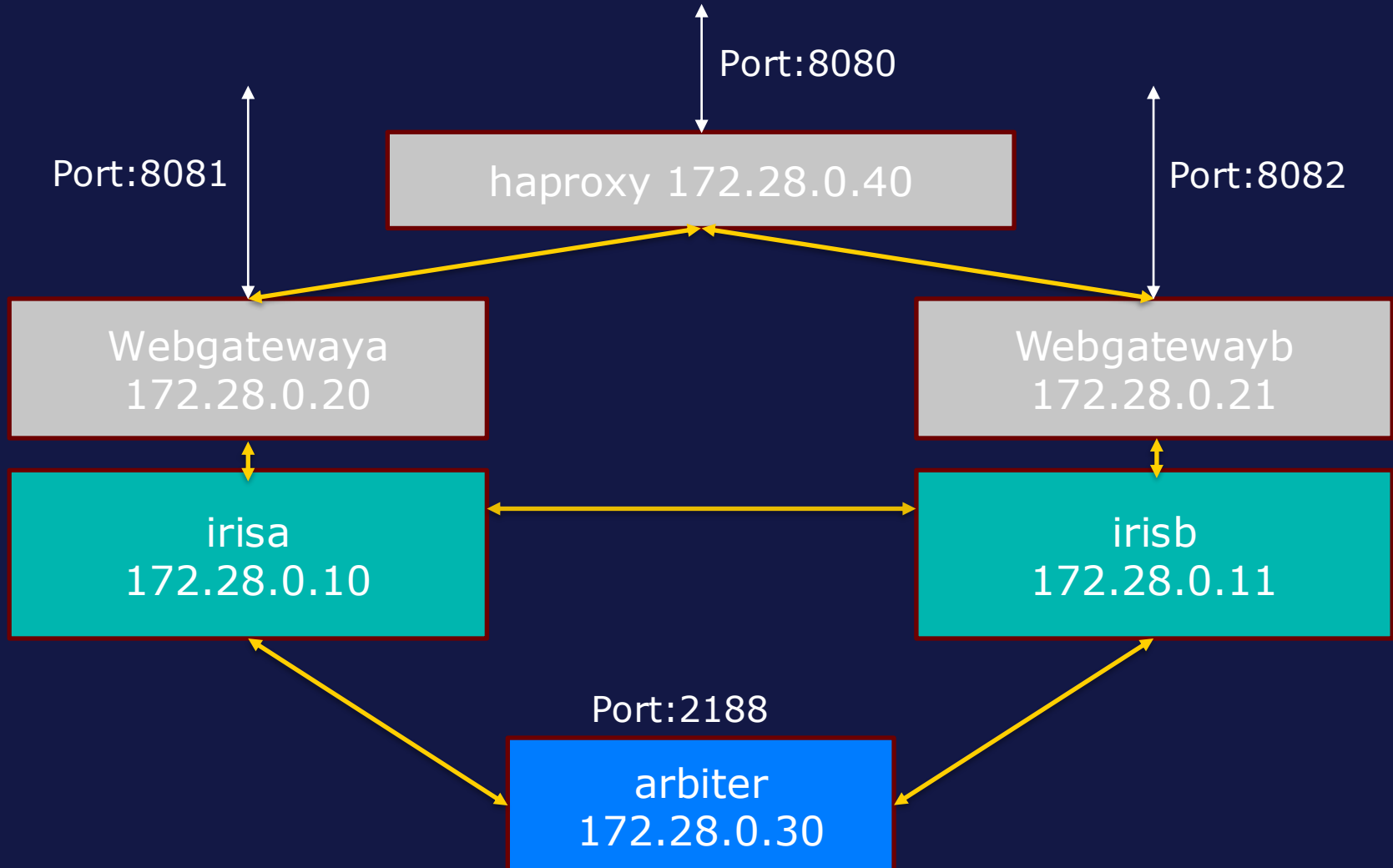
```
*** Recovery started at Fri Mar 20 15:10:18 2026
Current default directory: /Host/SYS/mgr
Log file directory: /Host/SYS/mgr/
WIJ file spec: /Host/SYS/mgr/IRIS.WIJ
Recovering local (/Host/SYS/mgr/IRIS.WIJ) image journal file...
Starting WIJ recovery for '/Host/SYS/mgr/IRIS.WIJ'.
  0 blocks pending in this WIJ.
Exiting with status 3 (Success)
03/20/26-15:10:18:801 (13254) 0 [Generic.Event] InterSystems IRIS license file (iris.key) must be validated with License Server.
03/20/26-15:10:18:918 (13254) 0 [Generic.Event] Allocated 1532MB shared memory
03/20/26-15:10:18:919 (13254) 0 [Generic.Event] 1024MB global buffers, 300MB routine buffers, 64MB journal buffers, 78MB buffer descrip
03/20/26-15:10:18:920 (13254) 0 [SIMD] SIMD optimization level: DEFAULT
03/20/26-15:10:18:954 (13254) 0 [WriteDaemon.UsingWIJFile] Using WIJ file: /Host/SYS/mgr/IRIS.WIJ
03/20/26-15:10:18:955 (13254) 0 [Generic.Event] Jrn info from prior WIJ (imflags: 0):
    wdpass: 0
    jrnwdpass: 2099
    fspec: /Host/SYS/mgr/journal/20260319.005
    filecnt: 1377
    fileoff: 0
    prevfcnt: 1377
    prevfileoeff: 0
    min trans cnt: 1377
    min trans index: 426784
03/20/26-15:10:18:963 (13254) 0 [Generic.Event]
Startup of InterSystems IRIS [IRIS for UNIX (Ubuntu Server LTS for ARM64 Containers) 2025.3 (Build 226U) Thu Nov 13 2025 12:40:36 EST]
in /usr/irissys/bin/
with mgr: /Host/SYS/mgr
with wij: /Host/SYS/mgr/IRIS.WIJ
from: /Host/SYS/mgr/
OS=[Linux], version=[#1 SMP Mon Feb 16 11:19:06 UTC 2026], release=[6.12.69-linuxkit], machine=[aarch64]
nodename=[CDH].
numasyncwijbuf: 2, wdwrite_asyncio_max: 64, wijdirectio: on, synctype: 3
System Initialized.
03/20/26-15:10:18:963 (13295) 0 [WriteDaemon.Started] Write daemon started.
```



Hands-on



Training Environment – docker based



Hands-on – Buildup training env

- Get the training env setup folder
- Put the folder on your target disk
- Buildup the env
 - Windows:
 - `.\Setup.ps1`
 - MacOS:
 - `bash setup.sh`
 - It will pull container images & setup automatically
- Check
 - Open in browser:
 - <http://127.0.0.1:8081/csp/sys/Utilhome.csp>
 - <http://127.0.0.1:8082/csp/sys/Utilhome.csp>
 - <http://127.0.0.1:8080/csp/sys/Utilhome.csp>



Controlling Instance From OS Command Line

- `iris` command provides variety of options to control instance.
 - On Windows, must run `iris` command from *InstallDirectory/bin* or set PATH.
 - On UNIX®/Linux, can run `iris` command anywhere.
- `iris help` displays all options.
 - `iris help display` (Windows) displays full help in browser.
 - `iris help option` (UNIX®/Linux) displays help for specific option.



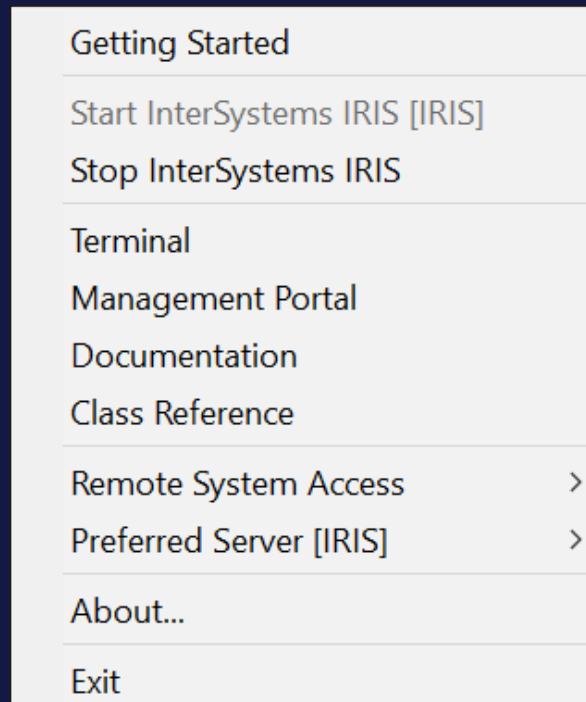
Start/Stop/Restart from OS Command Line

List all instances, including running status	<code>iris list</code>
Start instance	<code>iris start <i>InstanceName</i></code>
Stop instance	<code>iris stop <i>InstanceName</i></code>
Restart instance	<code>iris restart <i>InstanceName</i></code>



Start/Stop with InterSystems Launcher

- InterSystems Launcher installed on Windows.
 - InterSystems Launcher > Start/Stop.



Privileges to Start Instance

- UNIX® and Linux users must be one of the following:
 - User with sudo privileges.
 - Belong to GID group given start/stop privileges.
 - During installation can specify groups with privileges.



InterSystems Windows Terminal

- Provides ObjectScript shell.
 - InterSystems Launcher > Terminal.
 - From Windows OS:
`iris terminal instance`
- **Note:** does not use modern copy and paste keyboard shortcuts. Instead:
 - Use right-click to copy/paste.
 - Use alternate keyboard shortcuts:
 - Copy: **Ctrl+Insert** (not **Ctrl+C**).
 - Paste: **Shift+Insert** (not **Ctrl+V**).



Launch ObjectScript Shell Using `irissession`

- ObjectScript shell available using any terminal software.
 - To launch: `irissession instance`
 - Shell prompt displays current namespace.
`USER>`
- `irissession` can have additional arguments:
 - Namespace.
 - Method or routine to run.
 - Example:
 - `irissession instance -U namespace "method"`
 - `irissession instance -U namespace routine`



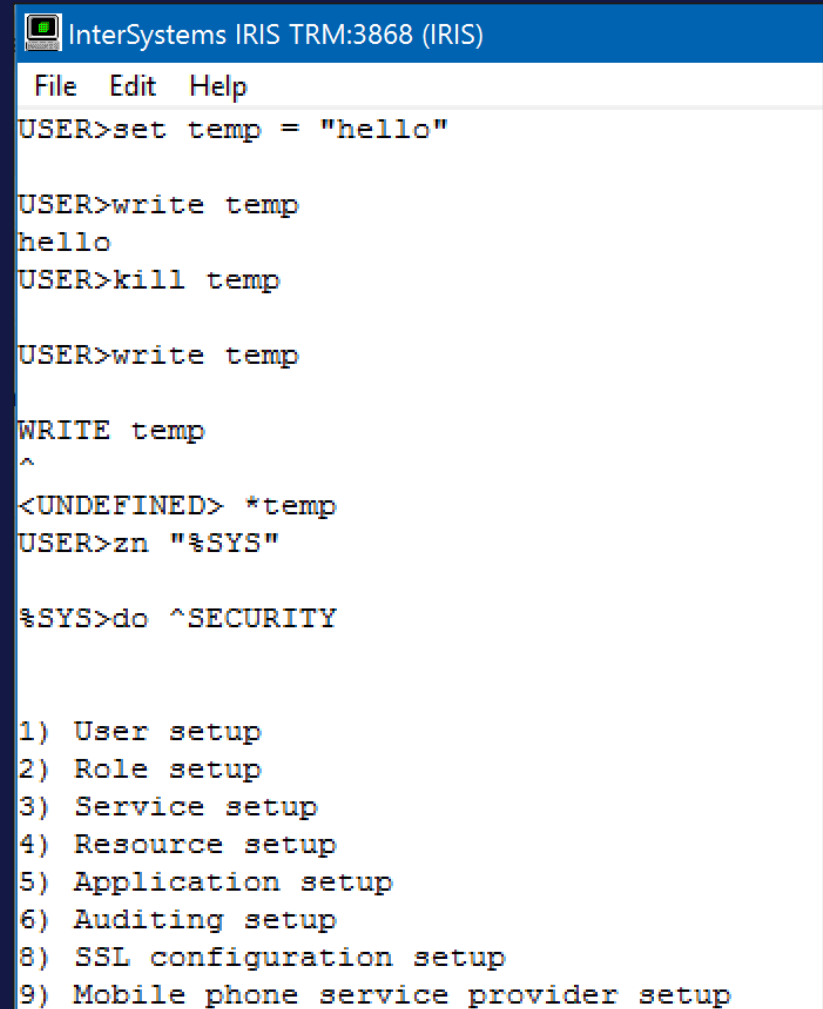
Launch ObjectScript Shell Using `irisession` (cont.)

- `irisession` with arguments used in OS scripts for automation.
- For more details:
 - Windows: `irisession help`
 - UNIX®/Linux: `iris help session`
- Alternate launch syntax (UNIX®/Linux):
 - `iris session instance`
 - `iris terminal instance`



Using ObjectScript Shell

- Changing namespace:
 - zn *"Namespace"*
 - do ^%CD
 - set \$namespace = *"Namespace"*
- Exiting Terminal:
 - halt
- Appendix A lists useful utilities.



```
InterSystems IRIS TRM:3868 (IRIS)
File Edit Help
USER>set temp = "hello"

USER>write temp
hello
USER>kill temp

USER>write temp

WRITE temp
^
<UNDEFINED> *temp
USER>zn "%SYS"

%SYS>do ^SECURITY

1) User setup
2) Role setup
3) Service setup
4) Resource setup
5) Application setup
6) Auditing setup
8) SSL configuration setup
9) Mobile phone service provider setup
```



Calling Utilities

- Some utilities called using legacy *^routine* syntax.
 - `%SYS>do ^DATABASE`
 - Utilities starting with `%` run in any namespace.
 - Utilities not starting with `%` run in `%SYS` namespace only.
- Some utilities called using `##class(package.class)` or `$SYSTEM.class` syntax.
 - `%SYS>do ##class(Backup.General).ExternalFreeze()`
 - Utilities starting with `%package` or `$system` run in any namespace.
 - Utilities not starting with `%package` run in `%SYS` namespace only.



Calling Utilities (cont.)

Syntax	Examples
<i>^ROUTINE</i> <i>TAG^ROUTINE</i>	do ^DATABASE, do ^CKEY do PURGE^JOURNAL
<i>^%ROUTINE</i> <i>^%SYS.ROUTINE</i> <i>TAG^%ROUTINE</i>	do ^%G, do ^%D, do ^%SS, do ^%CD do ^%SYS.GD do ALL^%FREECNT
<i>##class(Package.Class).Method()</i>	do ##class(Backup.General). ExternalFreeze()
<i>##class(%Package.Class).Method()</i>	do ##class(%SYS.Journal.File).Purge()
<i>\$SYSTEM.Class.Method()</i> Shorthand for: <i>##class(%SYSTEM.Class).Method()</i>	do \$SYSTEM.License.Upgrade() do \$SYSTEM.License.CKEY() write \$SYSTEM.License.ConnectionCount()



Operating System Shell

- Enter ! or \$ at the command line to invoke OS Shell.
- Enter *!command* or *\$command* at the command line to run OS command.

```
USER>!ipconfig

Windows IP Configuration

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : localdomain
    Link-local IPv6 Address . . . . . : fe80::6460:f7df:9183:7966%8
    IPv4 Address. . . . . : 172.16.239.140
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 172.16.239.2

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
```



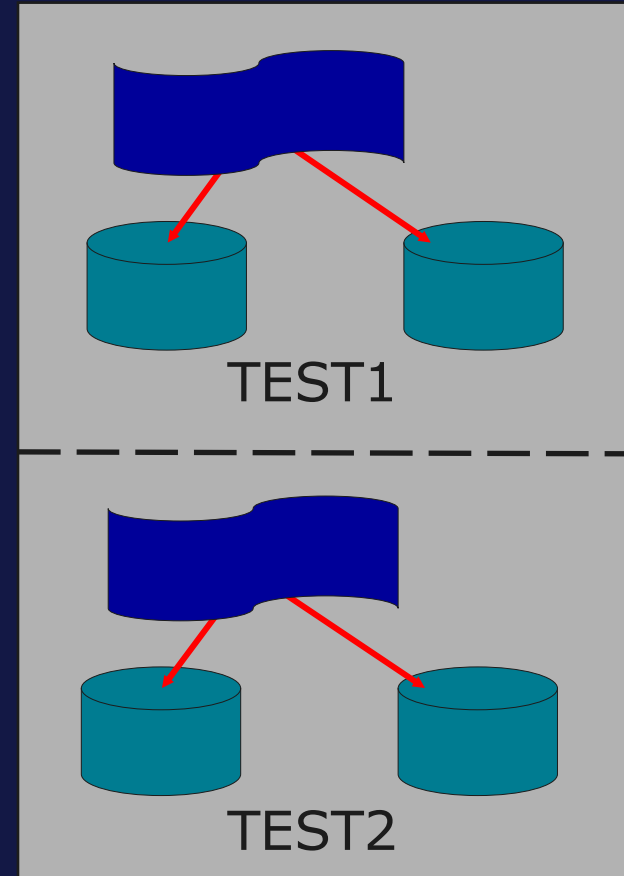
Forcing a Shutdown

- Used if instance hangs during shutdown.
 - `iris force InstanceName`
- Do not use in place of standard shutdown procedure.
 - Possible application data loss.
- Contact InterSystems Worldwide Response Center before forcing shutdown.



Multiple Instances

- Multiple instances can run concurrently on same server.
 - Each instance must have different name and SuperServer port.
- Refer to *Special Considerations Section* of appropriate OS in *Installation Guide* for instance details.



List Instances and Versions

```
Instance 'IRIS1'      (Development installation)
  directory: 'C:\InterSystems\IRIS'
  versionid: '2018.1.1.611.0'
  conf file: 'c:\intersystems\iris\iris.cpf' (SuperServer port = 51773,
WebServer port = 52773)
  status   : 'running, since Fri Mar 30 13:27:14 2018'
  Product  : InterSystems IRIS

Instance 'IRIS2'      (Development installation)
  directory: 'C:\InterSystems\IRIS1'
  versionid: '2018.1.1.611.0'
  conf file: 'c:\intersystems\iris1\iris.cpf' (SuperServer port = 51774,
WebServer port = 52774)
  status   : 'down, since Wed Apr 04 18:24:15 2018'
  Product  : InterSystems IRIS
```

- `iris list`
 - Lists information about each instance on machine.
 - Run from OS command line.
- `do ^VERSION or write $zversion`
 - Provides version of instance.
 - Run from ObjectScript shell.



Suggested Readings

- *Installation Guide.*
- *System Administration.*
 - *Managing on Windows.*
 - *Managing on UNIX[®], Linux, and Mac OS.*



Summary

- What are the key points for this module?

